



A SHORT GUIDE TO LIVING ROOFS

Composed Ecology

One roof, planted in May, in flower by September.

N A T I R A

Not green. Alive.

A roof can hold far more than most people are shown.

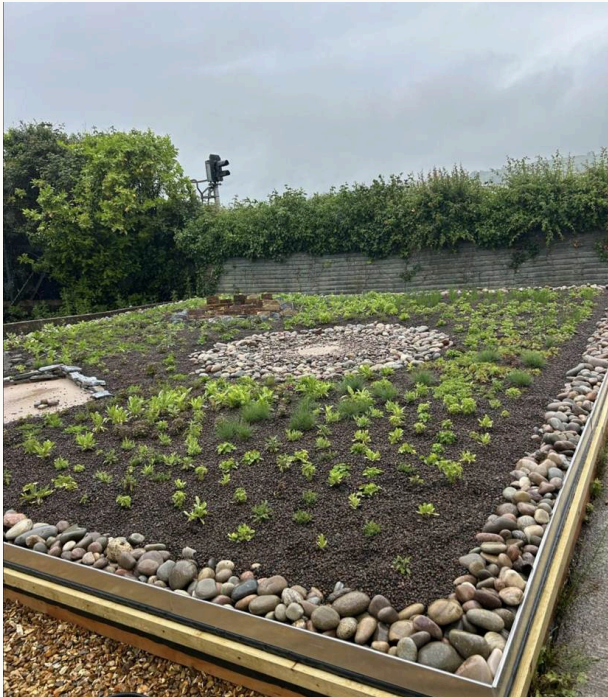
Most people come to green roofs through one word: sedum. It is a good plant and a sensible place to start. This guide goes further, and explains the way we plant roofs at Natira, which we call composed ecology.

We have grown living roofs on the shores of Larne Lough since 2008. Everything here comes from roofs we have planted and watched grow, including the display roof in these photographs, planted in May 2026 and followed through its first summer. It takes about ten minutes to read.



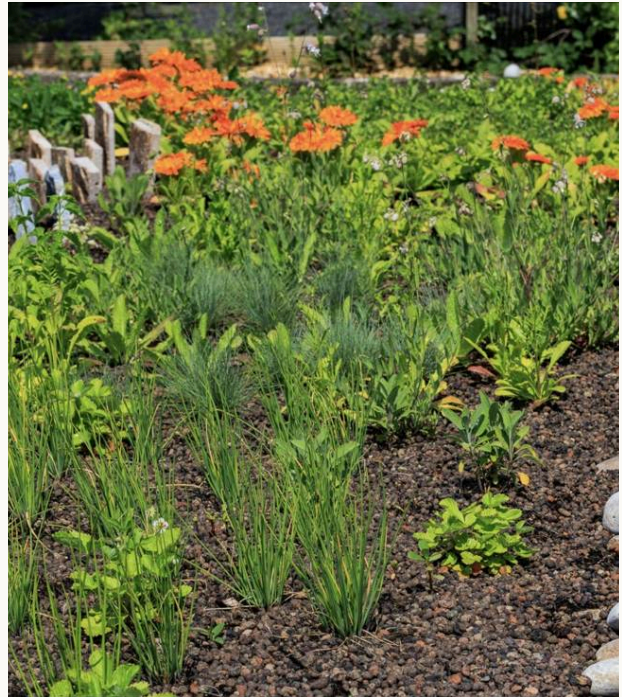
The display roof at Larne Lough two months after planting, with the sand bank and rubble reef in the foreground.

One roof through its first summer.



FOUR WEEKS · 1 JUNE 2026

Small plants in open substrate. It looks newly planted, because it is.



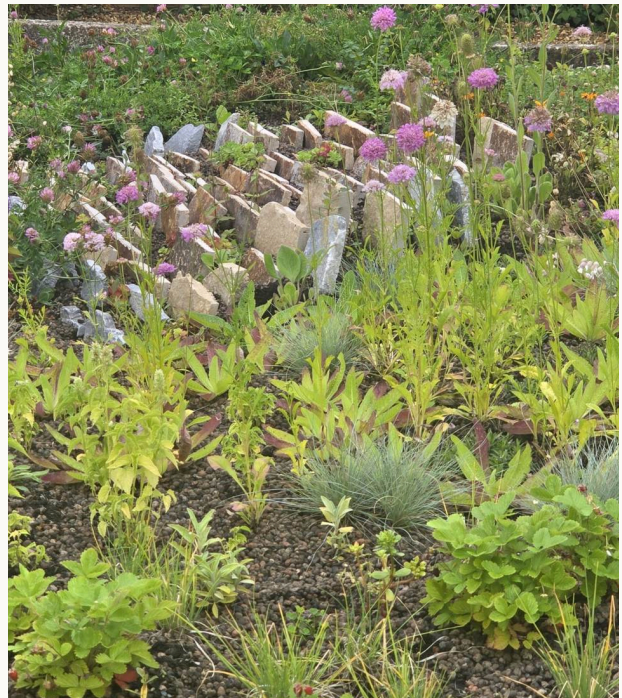
TWO MONTHS

Calendula in flower, chives filling out, the first gaps closing.



THREE MONTHS

Bird's-foot trefoil and borage in full flower. The substrate is disappearing.



FOUR MONTHS

Scabious in flower, and the roof reads as a meadow.

It arrives young, and it keeps getting better.

A planted roof begins at its smallest. Everything after that is growth.

PLANTING DAY

Small plants in open substrate, roughly 20cm apart.

YEAR ONE

Roots go down and plants spread. The fastest flower in the first summer. Watering and weeding matter most now.

YEAR TWO

The canopy knits together and the main flowering begins. The roof starts to look after itself.

YEAR THREE ON

The roof settles and seeds itself into its own gaps. Some plants move around. That is the roof working.

How the planting ages

Annuals

Calendula and borage flower in the first summer, set seed, and come back the next year wherever they choose.

Biennials

Teasel and foxglove spend their first year as a leafy rosette and flower in their second summer. The teasel planted on our roof in May 2026 will flower in 2027.

The lasting carpet

Thyme, bird's-foot trefoil and fescue spread into a low permanent cover. Scabious and yarrow give it height and structure.

Winter

Seed heads of teasel, scabious and yarrow stand above straw-coloured grasses, and the stones hold the shape of the roof until the late-winter cut.

Worth knowing. A planted roof asks for patience in its first few months. If a roof has to look complete the day the scaffolding comes down, say so at the start, and the planting can be planned around it.

The layers underneath are the same. What arrives on the lorry is not.

Pre-grown blanket

Sedum blanket is grown in a nursery for a year or more on a mat, then cut, rolled and delivered like turf. It is laid over the substrate in rows, and the roof is fully covered on the day of installation.

It carries a mix of sedums, low succulent plants that store water in their leaves and cope well with dry, exposed roofs. It is the most common green roof in Britain and Ireland, and we grow a great deal of it ourselves.

Plug planting

Plugs are young plants grown individually in small cells, each with its own root ball. They are planted by hand into the substrate, around 25 to every square metre, following a planting plan drawn for the roof.

Because each plant is placed one at a time, the roof can hold a far wider mix: wildflowers, herbs, grasses, coastal plants, and sedums too. This is how every composed ecology roof is planted.



Plug planting four weeks in. Each plant was placed by hand.

Where the care goes. A blanket arrives finished, and its care goes into keeping it that way. A feed each spring and one or two maintenance visits a year are typical.

A planted roof arrives young, and its care goes into the first year while it grows in. After that it asks for less each year. Both are proper green roofs: both manage rainwater, insulate, and protect the membrane underneath.

38

species on the 50 square metres of our display roof,
planted as 1,257 individual plugs.

A roof composed for the place it stands.

A composed roof is planned the way a good garden or a piece of music is planned: every part chosen for its place and its moment, and all of it working together. The idea rests on five principles.

01 Drawn from the landscape

The planting palette comes from the ground the building stands on. Basalt grassland on the Antrim coast and machair on a Donegal shore suggest two different roofs. Around Larne Lough the wild plants include thrift, sea campion, harebell and knapweed, and a roof here can draw on them.

02 Planted in groups

Plants that share the same depth, light and moisture are grouped together, the way they would grow in the wild. Each group feeds a different part of the roof's wildlife.

03 Composed across the seasons

The plan is set so something is in flower from early spring into autumn, with seed heads left standing through winter for birds and insects.

04 Built with habitat

Stone, sand and rubble are shaped into features that give insects places to nest, shelter, bask and drink.

05 Labelled honestly

Every plant is marked as a native or a garden guest, so you know exactly what is on your roof and why it is there.



Herbs, grasses and bird's-foot trefoil knitting together, three months after planting.

Each group has a job.

These are the six groups we plant most often. Some follow a wild plant community closely. Others, such as the Kitchen Garden and the Night Garden, are built around a purpose. On a commissioned roof they are a starting point, adjusted to your depth, your aspect and the landscape around you.

Wildflower Meadow

Scabious, yarrow, herb burnet, centaurea*, borage*, calendula*, bergamot*

Open, nectar-rich flowers for bees and hoverflies through midsummer.

Pollinator Corridor

Creeping thyme, marjoram, kidney vetch, bird's-foot trefoil, hyssop*, aubrietia*

A low flowering carpet. Bird's-foot trefoil is the main food plant of common blue caterpillars.

Kitchen Garden

Chives, oregano, alpine strawberry, sage*, garlic chives*, purslane*

A roof you can pick from. Chive and oregano flowers are busy with bees.

Seasonal Drift

Teasel, bladder campion, aquilegia, scabious, dianthus*, blue fescue*

Structure and movement. Teasel heads stand through winter, and goldfinches feed on the seed.

Night Garden

Bladder campion, sea campion, dianthus*, alyssum*

Pale flowers that show up at dusk. The campions are visited by night-flying moths.

Woodland Edge

Foxglove, red campion, aquilegia, campanula*, winter purslane*

For shaded roofs beside trees or taller buildings. Foxgloves are a favourite of bumblebees.

* Garden guest: a plant from outside Britain and Ireland, chosen for its flowers, its scent or its use in the kitchen.

Sedum has a place here too. We grow Sedum album, reflexum and anglicum as plugs. They sit in the driest parts of a roof and in the gaps between stones, alongside everything else.

What the stones and sand are for.

Plants feed wildlife. Habitat features give it somewhere to live. Each one is built from natural stone, sand or rubble and planted around its edges.



A stone crevice on the display roof, with calendula behind.

Stone Crevice

Slivers of stone set on edge with graded gaps between them. Sedums and thyme root into the cracks. The stone holds the sun's warmth, and insects shelter in the gaps.

Rubble Reef

A low mound of broken stone, always set beside the sand bank. Beetles, spiders and overwintering insects find cover in the spaces between the pieces.

Sand Bank

A patch of open sand with a planted margin. Many solitary bees nest in bare ground, and a sunny sand bank gives them a place to dig.

Dew Bowl

A shallow ring of stones that holds water for a while after rain. Damp-loving plants grow around it, and insects come to drink.

If you think it might need water, it does. Water it.

That is the only watering rule we give. Every living roof needs a little care, and for a planted roof most of the work comes early.

FIRST SUMMER

Water in dry weather while the roots establish. Pull out any weeds that blow in, while they are small. A walk across the roof every couple of weeks is enough to keep an eye on it.

SECOND SUMMER

With hotter summers, the roof may want a little water in a long dry spell. Weeding becomes lighter as the plants close the gaps themselves.

EVERY YEAR AFTER

Leave the seed heads standing through winter. Then one tidy in late winter: cut back the old stems, take the clippings off the roof, and check the outlets and gravel borders are clear.

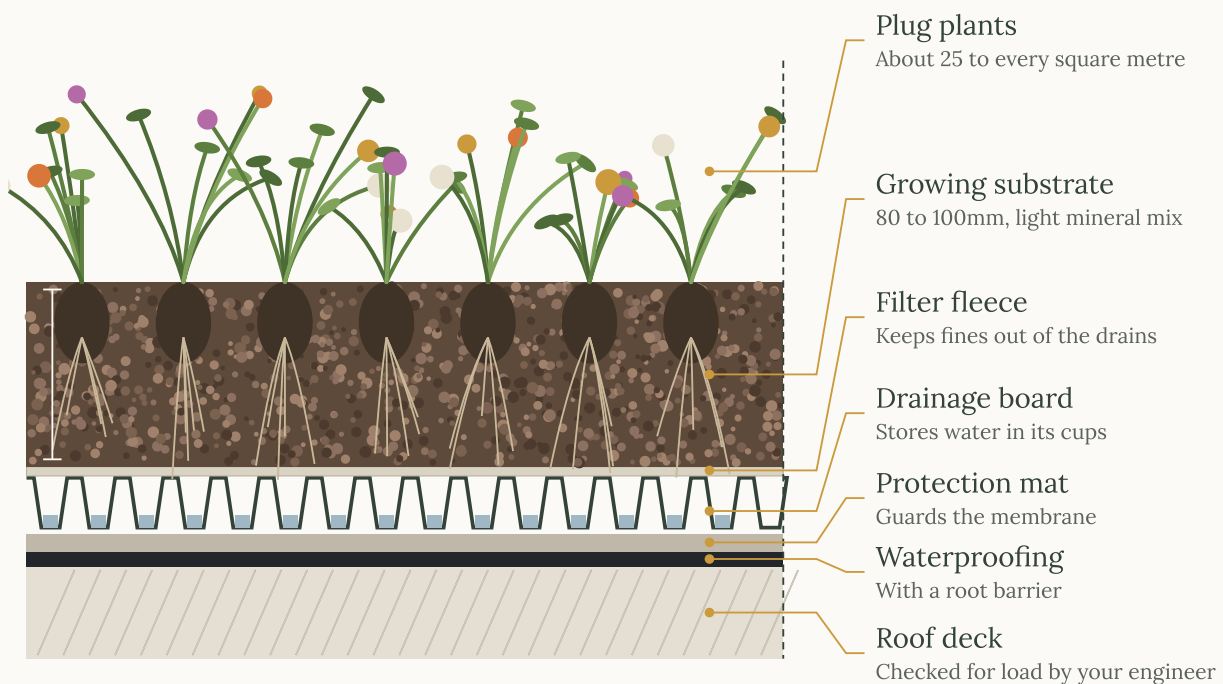
You will soon learn how your roof looks when it is thirsty and how it looks when it is happy, and you will stop needing the rule.



Chives and alpine strawberries at four weeks, the stage when watering matters most.

A garden, built in thin layers, sitting on your waterproofing.

Every green roof has the same basic build-up, whatever grows on top. Ours is around 100mm to 125mm deep in total, most of it growing substrate.



Section through a Natira roof. Not to scale.

Substrate, not soil

Garden soil is heavy when wet and packs down over time. A mineral substrate stays open, drains freely and keeps its depth for the life of the roof.

Water held in reserve

The cups in the drainage board fill after rain and give the roots something to draw on in a dry spell. The rest runs off to your outlets.

A root barrier

Roots look for water. A membrane with a root barrier keeps them in the garden and out of the waterproofing.

Borders kept clear

A gravel border around the edges and outlets keeps plants away from where water leaves the roof.

Weight. Our substrate weighs about 10kg per square metre for every 10mm of depth when fully soaked. At 100mm, the deepest we lay, that is around 100kg per square metre, plus about 20kg once the plants are established. The deepest parts of our display roof are 100mm, and thriving.

Substrates vary, so ask any supplier for the saturated weight of theirs. Your engineer should confirm your roof can carry the load.

Five questions to ask any green roof supplier.

Whoever you buy from, these answers will tell you what you are getting.

1 How deep is the substrate, and what does it weigh when saturated?
Ask for a figure in kilograms per square metre, fully soaked. Your engineer needs it, and it tells you what can grow.

2 What will the roof look like in year three?
Ask to see photographs of a roof they supplied, taken a few years after installation as well as on the day it went on.

3 Which plants are on it, and how many species?
A named list tells you what will flower, when, and what it will feed.

4 What care will it need, and who will do it?
Ask how often it needs feeding, weeding or watering, and whether that is something you will do or a visit you will pay for.

5 Where were the plants grown?
Plants grown in a similar climate to yours settle faster and last longer.

Send us your roof drawing.

Every Natira roof is drawn for the building it sits on. We are a small practice, and we grow every plant ourselves at Larne Lough. That is what lets us make each roof its own.

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- 01 Send us a drawing or photograph of your roof, with its size and location.
 - 02 We confirm the depth and loading with you or your engineer.
 - 03 We compose a planting plan from the groups that suit your place.
 - 04 Your plants are grown, delivered and planted, with a care guide for the years ahead.
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NATIRA

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